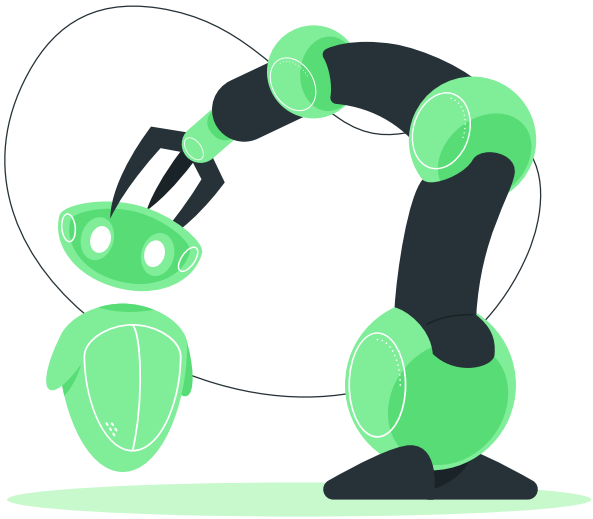
Robotics
CS 3015

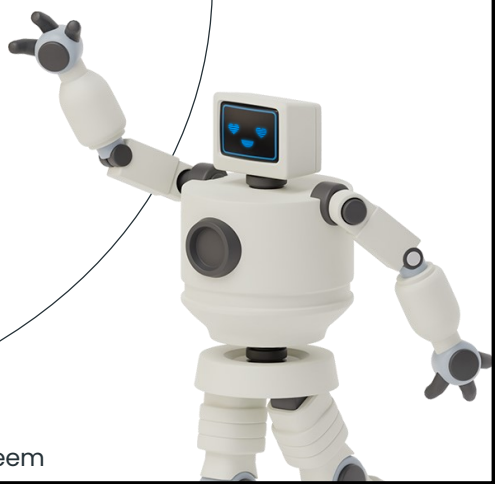
Umarah Qaseem



1

13

- Mobile Robots and Types
- Types of Robot Wheels
- Holonomic Systems
- Differential Drive Systems
- Ackerman Steering
- Forward Kinematics of Differential Drive Robot
- Inverse kinematics of wheel Robot
- The see think act cycle
- Obstacle Avoidance
- Control Logic



2

Mobile Robotics

-



Robotics – Umarah Qaseem

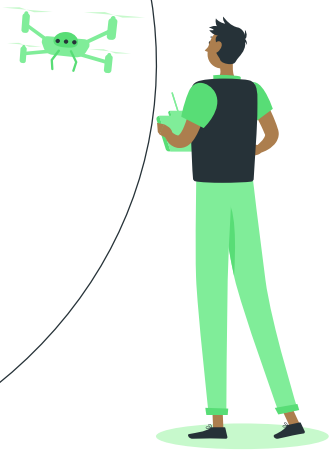
4

Mobile Robots

Robots that move!

Mobile Robotics

Study of Robots that move!



Robotics – Umarah Qaseem

5

Mobile Robot

- A mobile robot is a machine controlled by software that use sensors and other technology to identify its surroundings and move around its environment.
- Mobile robots function using a combination of artificial intelligence and physical robotic elements, such as wheels, tracks and legs.

6

Mobile Robot – Types

- Wheeled
- Legged
- Aerial
- Underwater
- Bio Inspired (Can also come under legged for some animals)
- Humanoid

7

Wheeled Robots





Robotics – Umarah Qaseem

8



9



10

Legged Robots





Robotics – Umarah Qaseem

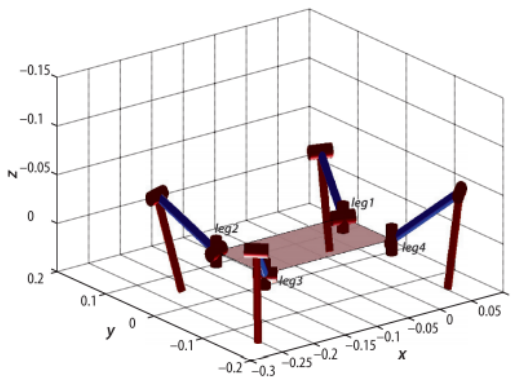
11

Legged Robots



Robotics – Umarah Qaseem

12





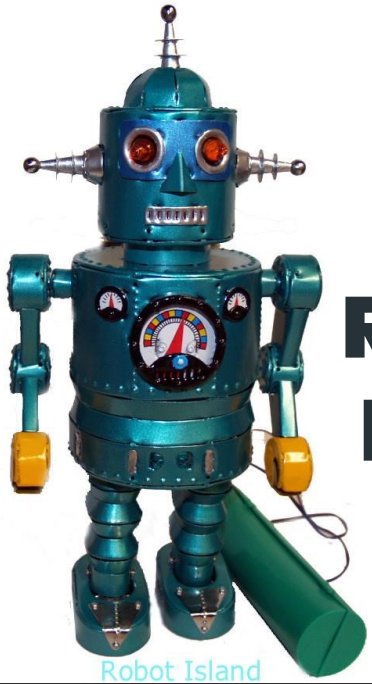
13

Humanoid Robots





14



Humanoid: Relate to your Robotic Arm?

15

Unmanned Aerial Ve





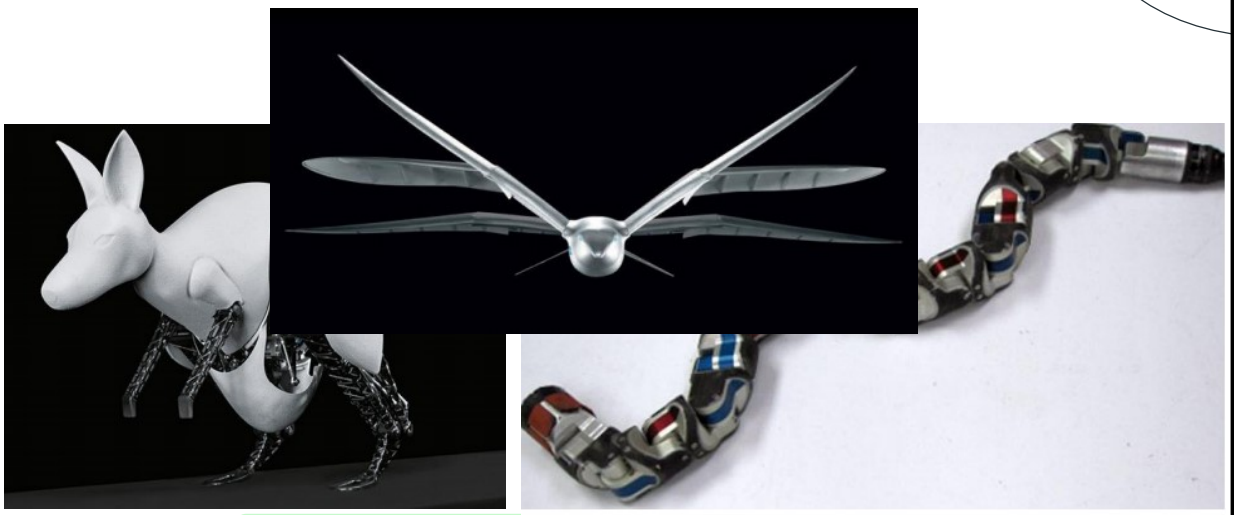
Autonomous Underwater Vehicle (AUV)



ed as UUV. Full form

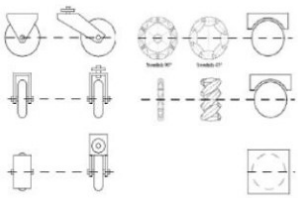


Bio-inspired Robot



18





Types of Wheels

19

Standard Fixed Wheel

- 2 degrees of freedom
- can move a robot backward and forward.

- Fixed wheels can be used for steering by using independently controlled wheel motors. They can also be attached to a rack and pinion system to steer a robot without using independently controlled motors



Orientable Wheels

- Orientable Wheels are generally used to stabilize robot. (e.g. grocery cart)
- Not powered

- **Two types**
- **Centered**
- **Off Centered**



Centered Oriented Wheel

- **Centered Oriented Wheel**: has a vertical axle that directly connects to the wheel.
- This wheel design is simple to integrate into most robot designs, and it delivers predictable motion characteristics along with static stability.



Robotics – Umarah Qaseem

22

Off Centered Oriented Wheel:

- The primary difference is that the axle's vertical axis is slightly off-centered.





Robotics – Umarah Qaseem

23

Orientable Wheel

- These wheels can also have 360-degree swivel joints that allow the wheel to rotate freely when a robot is in motion.



24

What is a castor wheel?

- Caster wheels are non-powered wheels used to provide extra contact points and keep your vehicle stable and enable them to turn around.

- Aka Off centered Oriented Wheel
- For the support of Robot

- Two types
- Fixed castor
- Swivel castor





25

Ball Wheel

- Also called ball castor
- are one of the simplest types of wheels used in robotics. These wheels are made up of either a metal or nylon ball mounted in a holder. The ball has 360-Degrees movement, and they are usually just used to balance a robot.

- Ball wheel's major disadvantage is that they have a high rolling resistance along with poor traction. They also don't do well on uneven terrain, and they are prone to jamming when used on dusty or dirty surfaces.



Robotics – Umarah Qaseem

26

Omni Wheels

- are the best type of robot wheel if you plan on building a robot that needs to quickly move in all directions. These wheels are made up of multiple roller wheels mounted around their circumference.
- The rollers on Omni wheels allow them to move forward, backward, and sideways. Omni wheels are also very efficient since they exhibit very low rolling resistance.



Robotics – Umarah Qaseem

27

Omni Wheels

- are also known as Swedish wheels and they can function as drive wheels, or they can be used to steer a robot.
- primary drawback is that they need to be oriented in a different direction than the drive wheels to allow the rollers to steer the robot.



Mecanum Wheels

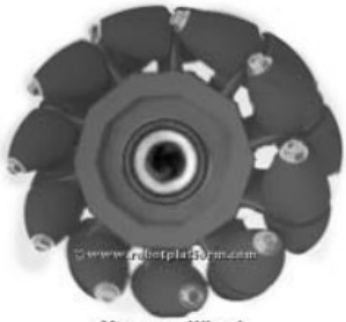
- are a type of Omni wheel that have rollers attached to the center spoke at a 45-degree angle. This configuration allows for faster dynamic motion in every direction.
- This is especially true for lateral motion since the rollers on Mecanum wheels are designed to quickly and efficiently transfer forward wheel spin into sideways motion.
- Mecanum Omni wheels also have the advantage of working for both driving and steering since the wheels can be mounted facing in the same direction.



Wheels



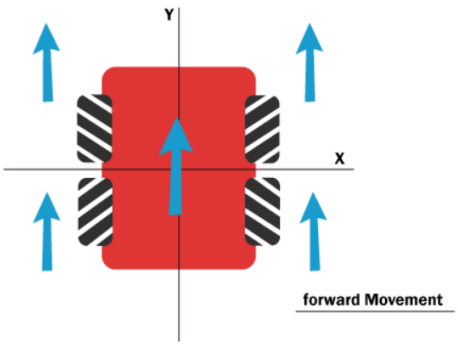
Omni Wheels



Mecanum Wheel

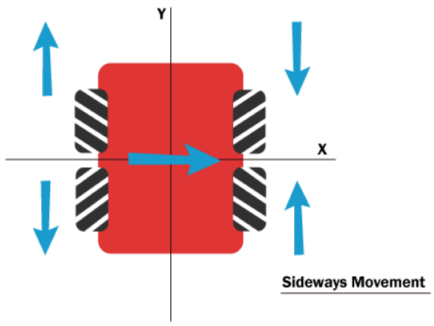
Mecanum – Forward / Backward

- When a four-wheel vehicle (in this case a robot) wants to move forward and backward, the Mecanums act like regular wheels. Their rollers move forward or backward to move the vehicle.



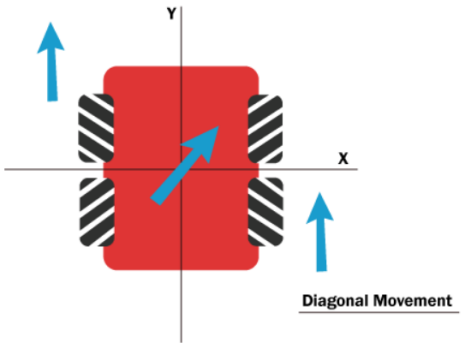
Mecanum – Sideways

- What happens when our four-wheel vehicle wants to move to the sides? In this case, as you see in the picture the two diagonally opposite Mecanum wheels move in one direction and the other two in the opposite direction. One of two the diagonally opposite wheels applies force to the vehicle, and the other one applies the driving force to the outside of the vehicle. The result will be a 180 degrees turn or side ward movement.

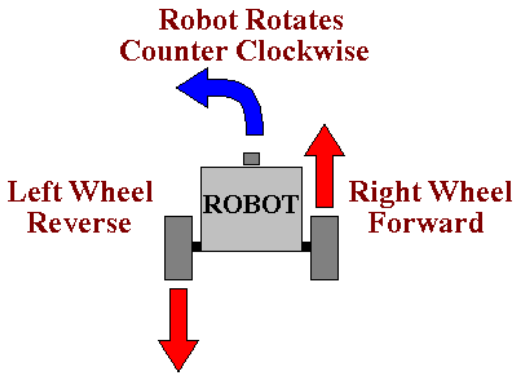


Mecanum – Diagonal

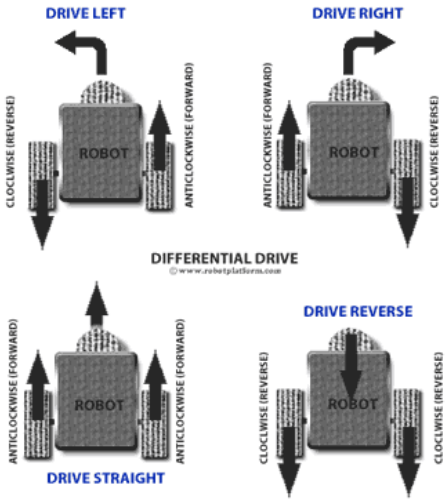
- To move our four-wheel vehicle diagonally, only one of the two diagonally opposite wheels needs tc move. As you see in the picture below, our vehicle moves diagonally with the help of only two wheels.



Fixed Wheel Robot Motion Cases



Fixed Wheel Movement



Robot Tracks

- are an excellent alternative to wheels for a robot that needs to navigate rough uneven terrain. Robot tracks also work well on soft surfaces like sand or snow since they evenly distribute the weight of a robot.
- Tracked robots also can climb up steep vertical grades, and they can even handle stairs with certain track configurations.





13b

Holonomic Systems



Holonomic System

- A system is **holonomic** if controllable degree of freedom is equal to total degrees of freedom.
- A holonomic robot can drive straight to a goal that is not in-line with its orientation

Non-Holonomic System

- If the controllable degree of freedom is less than the total degrees of freedom, then it is known as **non-Holonomic** drive.
- A non-holonomic robot must either rotate to the desired orientation before moving forward or rotate as it moves.

Non-Holonomic System

- System that have certain constraints over their velocity and movement
- Parallel parking problem is also due to non-holonomic constraints.

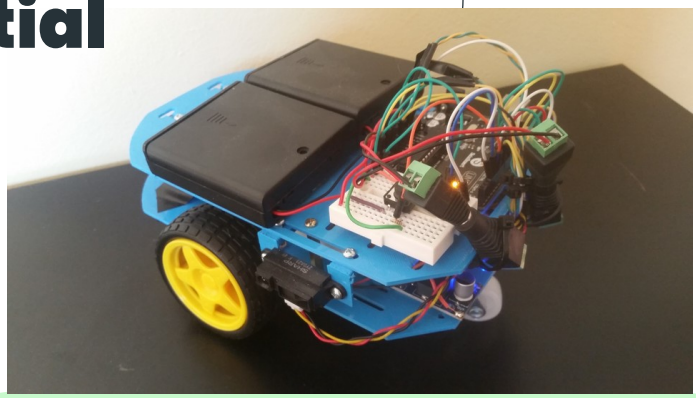
40

Example – Non-Holonomic

- A car has three degrees of freedom; i.e. its position in two axes and its orientation.
- However, there are only two controllable degrees of freedom which are acceleration/brake (translation in one axis) and turning angle of steering wheel (rotation).
- This makes it difficult for the driver to turn the car in any direction (unless the car skids or slides).

41

Differential Drive



43

Differential Drive

- Many mobile robots use a drive mechanism known as differential drive.
- It consists of 2 drive wheels mounted on a common axis.
- Each wheel can independently be driven either forward or backward
- Each wheel has independent actuators (motors)

44

Do regular cars have differential drive mechanism?

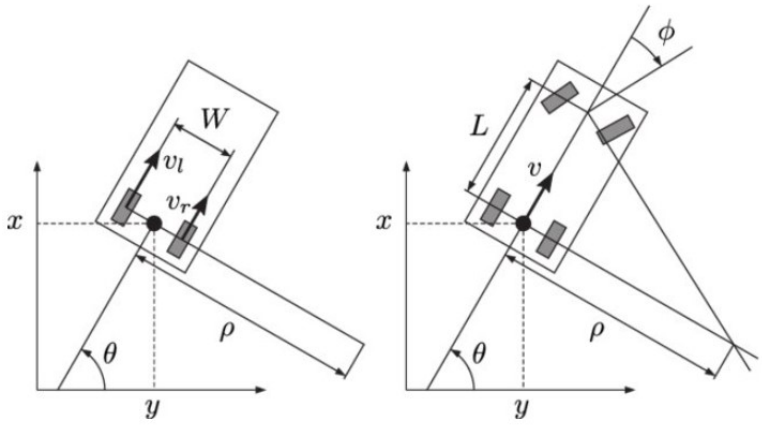
45

Ackerman Steering

- Like a regular Car!
- When a vehicle is turning, the inner front wheel needs to turn at a different angle to the outer because they are turning on different radii. The Ackermann steering mechanism is a geometric arrangement of linkages in the steering of a vehicle designed to turn the inner and outer wheels at the appropriate angles.

46

Differential Drive vs Ackerman



47

Differential Drive vs Ackerman

- So there are two types of drives.
- Differential drives (used in robots)
- Steered Drive (in our cars)
- Also known as ackerman steering

48

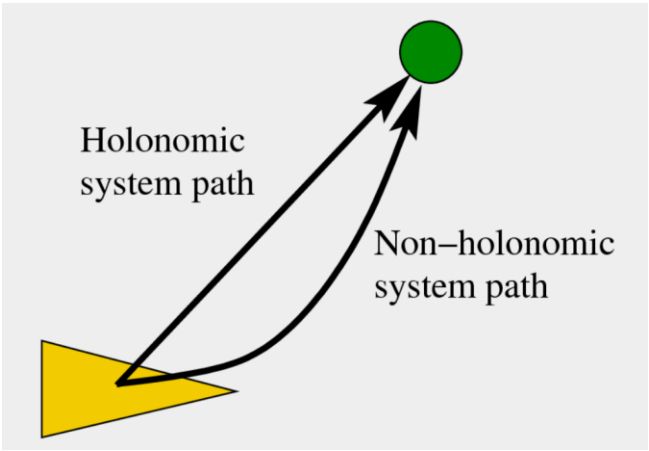
Are differential drive systems holonomic?

No!

50

Differential Drive holonomic? Why?

- For differential drive systems, the non-holonomic constraint means that the robot can not drive straight to a goal that is not in-line with its orientation. It must either rotate to the desired orientation before moving forward or rotate as it moves.



Robotics – Umarah Qaseem

51



The key Question!

52

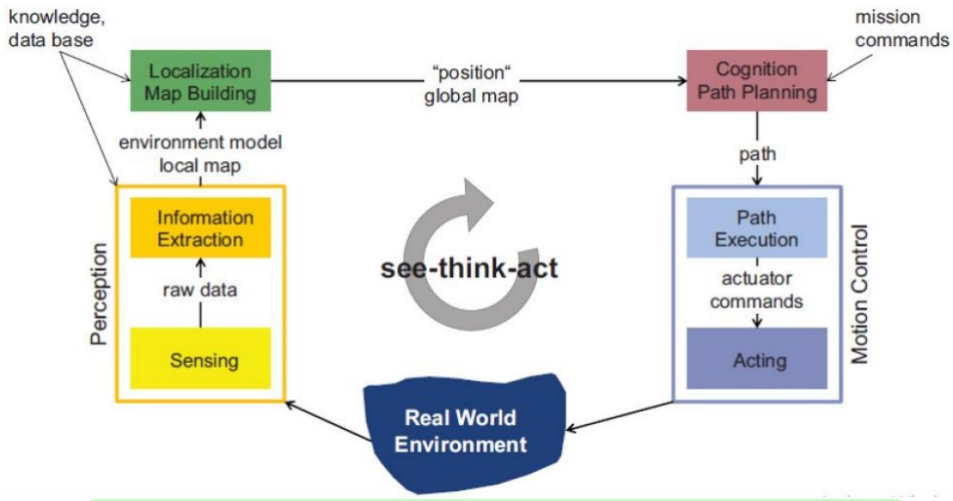
Autonomous Mobile Robot

- The three key questions in Mobile Robotics
  - Where am I ?
  - Where am I going ?
  - How do I get there ?
- To answer these questions the robot has to
  - have a model of the environment (given or autonomously built)
  - perceive and analyze the environment
  - find its position/situation within the environment
  - plan and execute the movement

Robotics – Umarah Qaseem

53

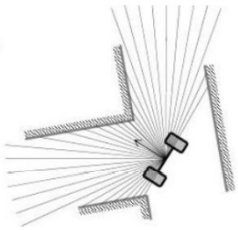
The see-think-act Cycle



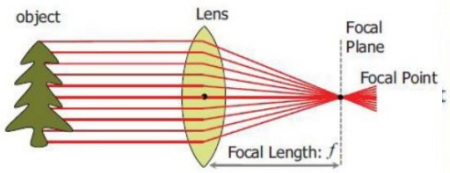
54

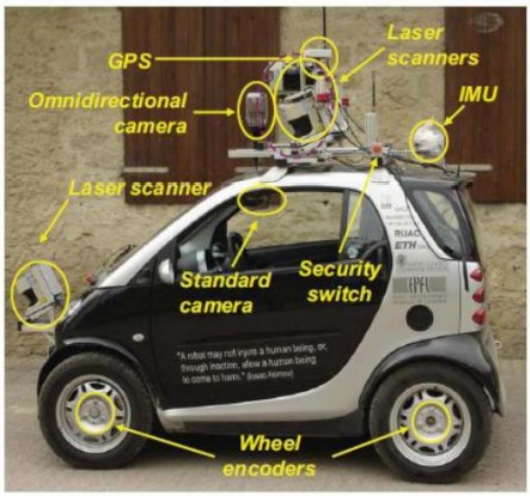
Perception (Sensing)

- Laser scanner
  - time of flight



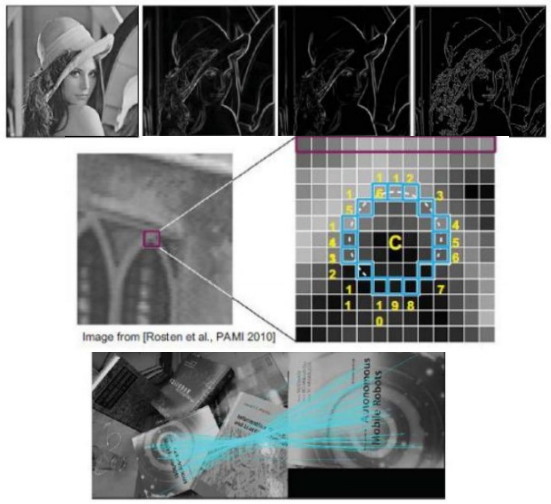
- Camers



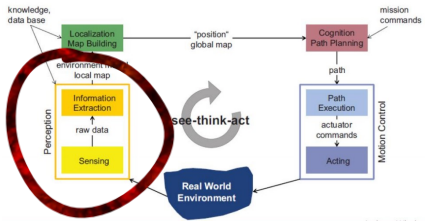


56

Perception (Sensing)

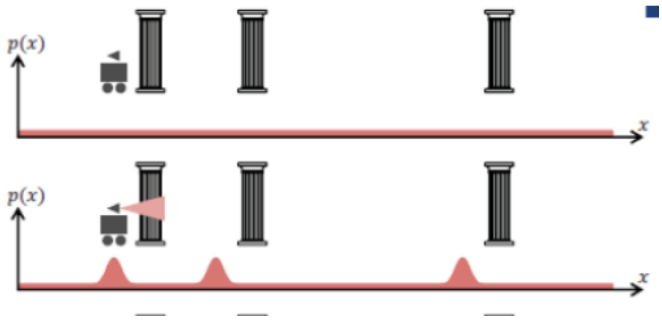


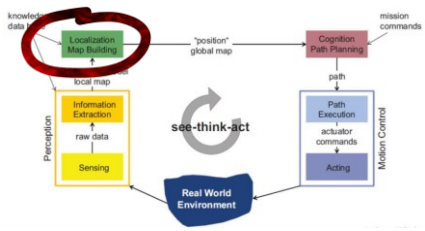
- Filtering / Edge Detection
- Keypoint Features
  - features that are reasonably invariant to rotation, scaling, viewpoint, illumination
  - FAST, SURF, SIFT, BRISK, …



Localization

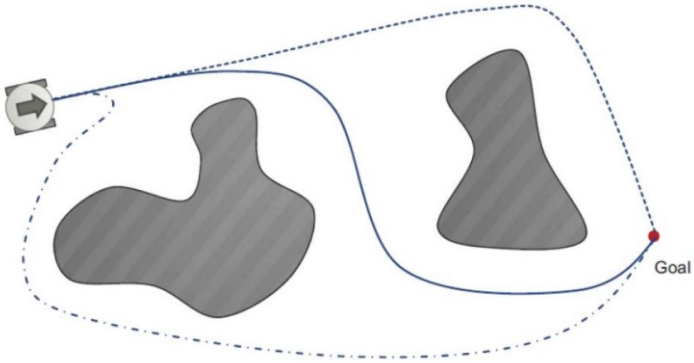
- Robot queries its sensors and find itself next to a pillar.

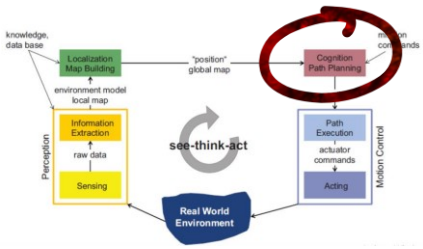




Cognition

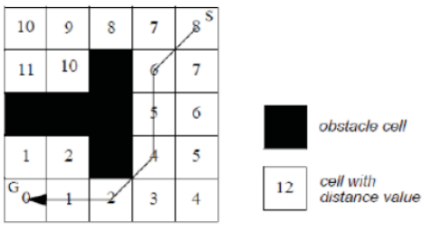
- Where am I going? How do I get there?



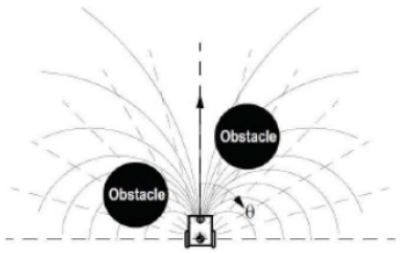


Path Planning

- Global Path Planning
  - Graph Search

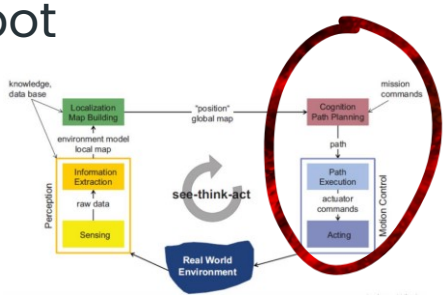


- Local Path Planning
  - Local Collision Avoidance



Motion Control

- Path Execution
- Actuators
- Motors
- E.g. Obstacle avoider robot
- Assignment?

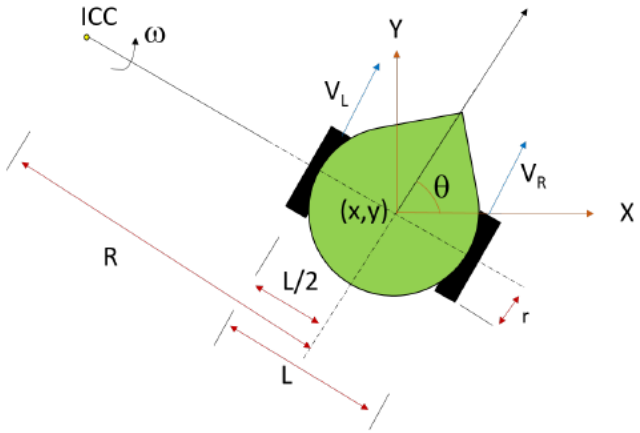


61

Mobile Robot Kinematics

62

Differential Drive Kinematics



64

Instantaneous Center of Curvature

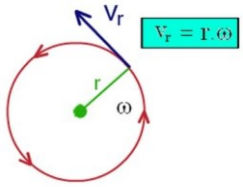
- While we can vary the velocity of each wheel, for the robot to form circle/rolling motion, the robot must rotate about a point that lies along their common left and right wheel axis. The point that the robot rotates about is known as the **ICC.**
- **Instantaneous Center of Curvature.**

65

Differential Drive Kinematics

- By varying the velocities of the two wheels, we can vary the trajectories that the robot takes.
- The rate of rotation ω about the ICC must be the same for both wheels

$$\omega\,(R + l/2) = V_r \quad (1)$$
$$\omega\,(R - l/2) = V_l \quad (2)$$



- where ***l*** is the distance between the centers of the two wheels, ***Vr ;***
- ***Vl*** are the right and left wheel velocities along the ground
- **R** is the distance from the ICC to the midpoint between the wheels

Differential Drive Kinematics

- The velocity of the robot is the average of the individual wheel velocities and is given by

$$V = \frac{V_r + V_l}{2}$$

Assumptions

- While modeling the robot,the following assumptions are made.

1. Robot is moving with constant velocity.

2. The duration of the robot motion is within a small period of time and so the velocities Vr and Vl would be constant in that duration.

3. The wheels of the robots do not slip and the surface for robot motion is flat.

68

R and ω

- At any instance in time, we can solve for R and ω.

$$R = \frac{l}{2}\frac{V_l + V_r}{V_r - V_l}; \quad \omega = \frac{V_r - V_l}{l}; \tag{3}$$

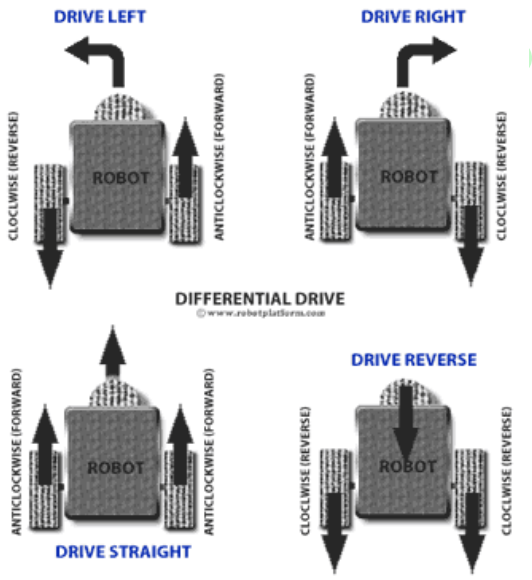
69

Wheel Robot Motion Cases

- There are four cases explaining the relation between velocities and direction.

- If *Vr* = *Vl*, robot will move **straight in linear direction**. *R* is infinite and ω Will be zero.
- If *Vr* = -*Vl*, robot will **rotate in the same place** along its axis about its center point. So *R* is zero.
- If *Vl* = 0, robot will **turn about left** wheel with the radius of *R* = *l*2.
- If *Vr* = 0, robot will **turn about right** wheel with a radius of *R* = *l*2.

Wheel R



Sensitivity to Velocity

- Differential drive vehicles are very sensitive to slight changes in velocity in each of the wheels.
- Small errors in the relative velocities between the wheels can affect the robot trajectory.
- They are also very sensitive to small variations in the ground plane, and may need extra wheels (castor wheels) for support

Forward Kinematics – differential drive

- Control Parameters: V_i and V_l
- Pose: (x', y', θ')

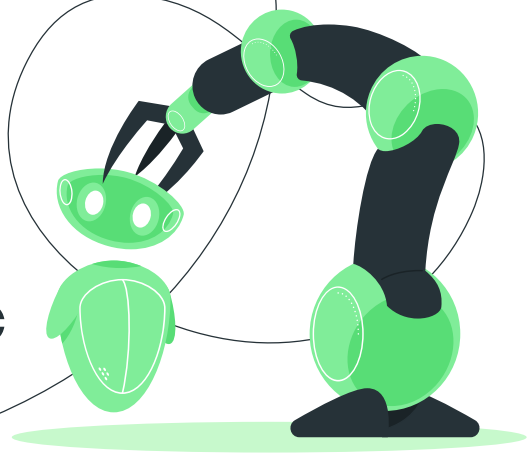
- What will be the pose or (x', y', θ') with the given mobile robot coordinates (V_i and V_l) - This is forward kinematics

Robot moves from (x, y, θ) to (x', y', θ')

Velocity information $u = (v, \omega)$

Point to Point/Goal Motion

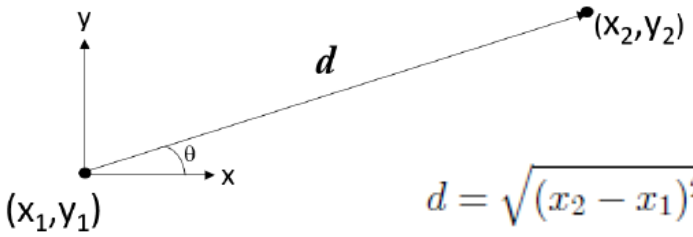
Control Logic



88

Control Logic

- For Point to point Motion



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

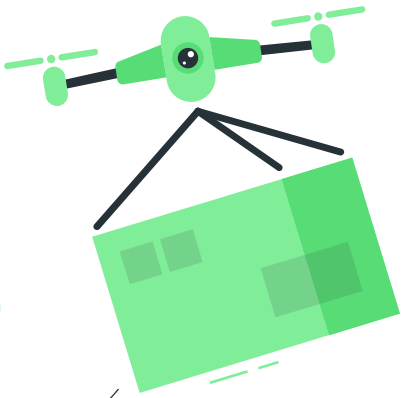
89

Control Logic

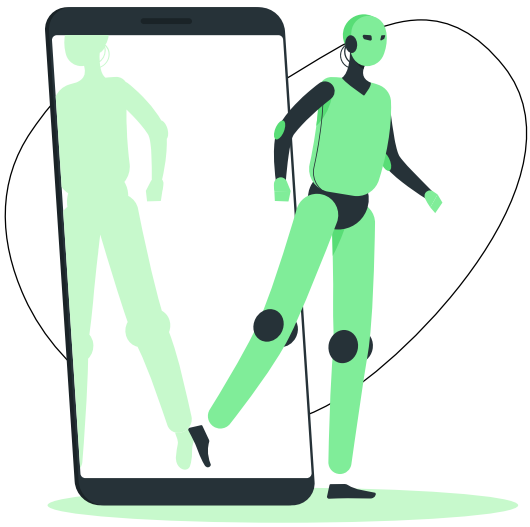
- With help of this formula, the velocity of the robot is decided. So long as,
- The distance is greater than 0, velocity is set to high and as the robot is at a distance of less than threshold (i.e. 10 cm from the destination), the velocity is lowered to zero to avoid overshoot.

90

Project Progress? Discussion



92



Thanks!

CREDITS: This presentation template was created by **Slidesgo**, including icons by **Flaticon**, infographics & images by **Freepik** and illustrations by **Stories**

100